SQL

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```sql
create database task1;
use task1;
CREATE TABLE Departments (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50)
);

CREATE TABLE Employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    dept_id INT,
    salary DECIMAL(10,2),
    manager_id INT,
    join_date DATE,
    FOREIGN KEY (dept_id) REFERENCES Departments(dept_id),
    FOREIGN KEY (manager_id) REFERENCES Employees(emp_id)
);
```

```sql

CREATE TABLE Projects (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(100),
    dept_id INT,
    budget DECIMAL(12,2),
    FOREIGN KEY (dept_id) REFERENCES Departments(dept_id)
);

CREATE TABLE EmployeeProjects (
    emp_id INT,
    project_id INT,
    hours_worked INT,
    PRIMARY KEY (emp_id, project_id),
    FOREIGN KEY (emp_id) REFERENCES Employees(emp_id),
    FOREIGN KEY (project_id) REFERENCES Projects(project_id)
);
```

```sql
CREATE TABLE Orders (
    order_id INT PRIMARY KEY,
    customer_id INT,
    order_date DATE,
    amount DECIMAL(10,2),
    status VARCHAR(20)
);

INSERT INTO Departments VALUES
(1, 'Engineering'),
(2, 'HR'),
(3, 'Sales'),
(4, 'Marketing'),
(5, 'Finance'),
(6, 'Research');
INSERT INTO Employees VALUES
(101, 'Aarav', 1, 90000, NULL, '2022-01-10'),
(102, 'Priya', 1, 70000, 101, '2022-03-15'),
(103, 'Rahul', 1, 70000, 101, '2023-02-20'),
(104, 'Sneha', 2, 60000, NULL, '2021-06-01'),
(105, 'Kiran', 2, 45000, 104, '2023-07-12'),
(106, 'Ananya', 3, 80000, NULL, '2020-11-05'),
(107, 'Vikram', 3, 50000, 106, '2024-01-18'),
(108, 'Meera', 4, 55000, NULL, '2022-09-25'),
(109, 'Arjun', 5, 75000, NULL, '2021-04-14'),
(110, 'Divya', 1, 50000, 101, '2024-05-30');
INSERT INTO Projects VALUES
(201, 'Website Redesign', 1, 500000),
(202, 'Mobile App', 1, 800000),
(203, 'Recruitment Portal', 2, 200000),
(204, 'Sales Dashboard', 3, 350000),
(205, 'Market Analysis', 4, 250000);
INSERT INTO EmployeeProjects VALUES
(101, 201, 120),
```

```sql
INSERT INTO EmployeeProjects VALUES
(101, 201, 120),
(102, 201, 100),
(103, 201, 100),
(102, 202, 150),
(103, 202, 130),
(110, 202, 90),
(104, 203, 80),
(105, 203, 110),
(106, 204, 140),
(107, 204, 100),
(108, 205, 160);
INSERT INTO Orders VALUES
(301, 1001, '2026-01-05', 15000, 'Completed'),
(302, 1002, '2026-01-10', 25000, 'Completed'),
(303, 1001, '2026-02-05', 20000, 'Completed'),
```

```sql
-- 1.Show department name, employee count, average salary, minimum salary, and maximum salary. Include departments with no employees.
select d.dept_name,count(e.emp_id),avg(e.salary),min(e.salary),max(e.salary) from Departments d left join Employees e on d.dept_id=e.dept_id group by d.dept_id,d.dept_name;

-- 2.Using a window function, return employee(s) earning the second distinct highest salary in each department.
select dept_id,emp_id,emp_name,salary from (select e.*, dense_rank() over (partition by dept_id order by salary desc) as salary_rank
from Employees e) x where salary_rank = 2;


-- 3.Find employees whose salary is greater than the average salary of their own department.
select*from employees e where e.salary>(select avg(e.salary) from employees where dept_id=e.dept_id);

-- 4.Find the employee(s) who earn the highest salary in each department. If multiple employees share the highest salary, return all of them.
select m.emp_id as manager_id,m.emp_name as manager_name,m.salary as manager_salary,count(e.emp_id) as direct_report_count,sum(e.salary) as team_salary from Employees m
join Employees e on e.manager_id=m.emp_id group by  m.emp_id,m.emp_name,m.salary having direct_report_count>=2;

-- 6.Find departments that have no matching project.
select d.dept_id,d.dept_name from Departments d left join Projects p ON d.dept_id=p.dept_id where p.project_id is null;
```